INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

MODULE-2

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PROJECT NAME: HotelRevAI AI Driven Revenue Analysis for Hotels_Batch -13

1. Objective

The goal of this module is to break down summary booking data into a daily level granular format. This allows for the analysis of daily revenue trends, occupancy rates per specific date, and the calculation of daily KPIs.

2. Creation of Excel Tables:

2.1 Creation of Duration_Details Table:

The **Bookings table** containing all booking details was used as the base dataset. From this table, selected columns such as BookingID, DateID, RoomID, CustomerID, CheckInDate, CheckOutDate, NightsStayed, and Total Price were extracted to create a new table named **Duration_Details**.

	A	B	C	D	E	F	G	H
1	BookingI	CustomerI	RoomTypeI	BranchI	CheckInDat	CheckOutDat	Duratio	Revenu
2	BK1000	C2587	RT9	B14	10/25/2023	10/27/2023	2	24000
3	BK1001	C2700	RT1	B1	5/28/2025	5/29/2025	1	2500
4	BK1002	C3158	RT10	B1	12/10/2025	12/12/2025	2	10000
5	BK1003	C2935	RT2	B7	1/24/2025	1/29/2025	5	20000
6	BK1004	C1997	RT6	B7	6/22/2024	6/24/2024	2	16000
7	BK1005	C4221	RT7	B14	5/28/2024	6/1/2024	4	7200
8	BK1006	C4894	RT3	B13	1/27/2026	1/28/2026	1	6000
9	BK1007	C456	RT8	B3	5/4/2024	5/9/2024	5	35000
10	BK1008	C3708	RT7	B11	8/22/2025	8/23/2025	1	1800
11	BK1009	C2101	RT10	B12	6/30/2024	7/7/2024	7	35000
12	BK1010	C4761	RT10	B7	11/27/2024	12/4/2024	7	35000
13	BK1011	C3709	RT9	B3	7/25/2025	7/28/2025	3	36000
14	BK1012	C4958	RT4	B1	9/27/2023	10/1/2023	4	36000
15	BK1013	C2753	RT3	B10	5/24/2025	5/31/2025	7	42000
16	BK1014	C2098	RT4	B5	3/26/2024	3/28/2024	2	18000
17	BK1015	C4254	RT3	B3	4/21/2024	4/26/2024	5	30000
18	BK1016	C2938	RT6	B4	3/21/2024	3/25/2024	4	32000
19	BK1017	C2976	RT11	B8	2/24/2025	3/3/2025	7	140000
20	BK1018	C2303	RT9	B5	[illegible]	[illegible]	6	72000

... | Branch | RoomBooked | RoomBookedCount | **Duration Details** | Date | TBI_Rates | TBI_Bookir

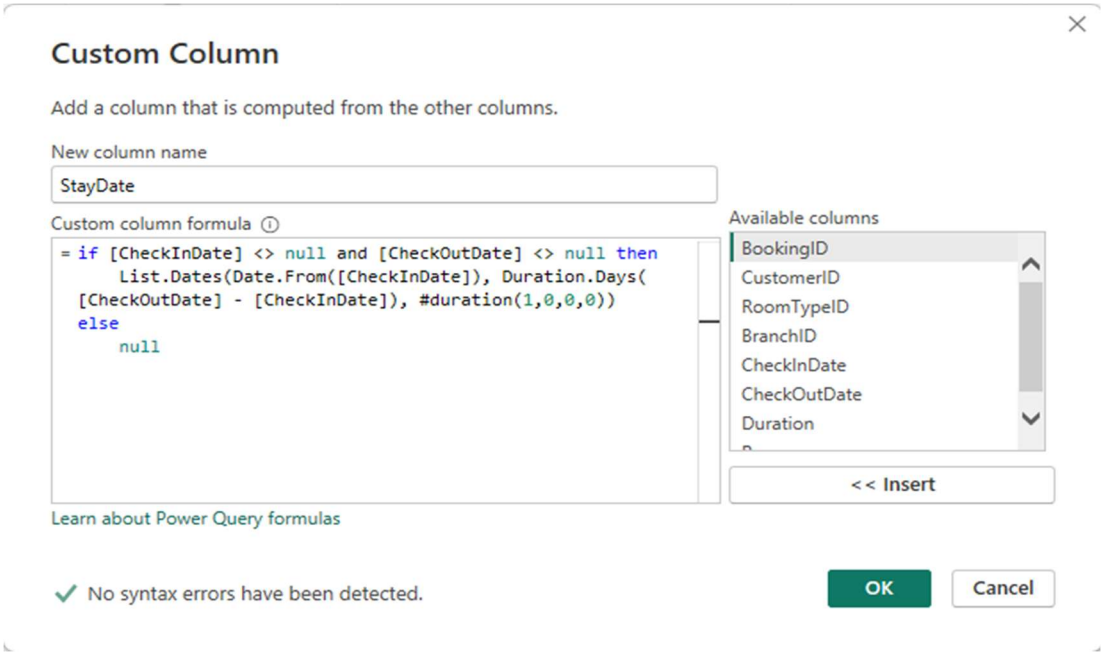
The table was selected and loaded into **Power Query** using **Data Tab → Get & Transform Data → From Table/Range** to begin data transformation.

2.1.1 Add Custom Column

In Power Query Editor, a custom column named StayDate was created using Add Column → Custom Column. The M-code formula generates a list of dates between the check-in and check-out dates for each booking

Formula:

```
if [CheckInDate] <> null and [CheckOutDate] <> null then
   List.Dates(Date.From([CheckInDate]), Duration.Days([CheckOutDate] - [CheckInDate]), #duration(1,0,0,0))
else
   null
```

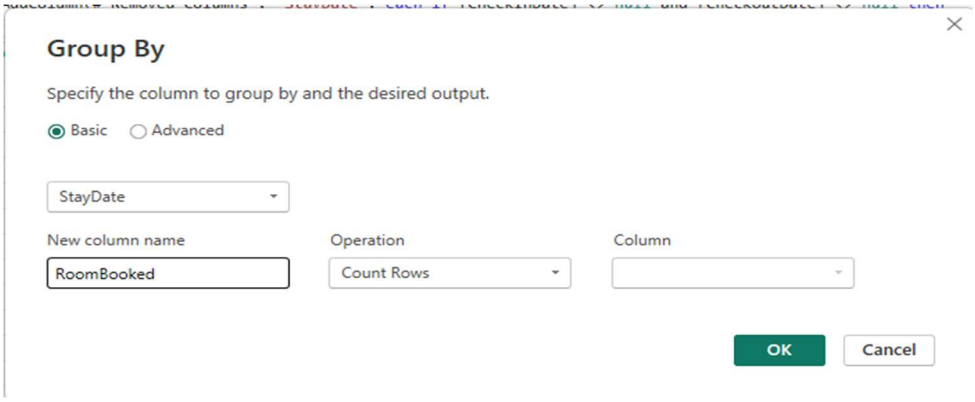


2.1.2 Expand Rows:

- Click the Expand arrows (two diverging arrows) next to the new StayDate column header.
- Select "Expand to New Rows".
- *Result:* Each booking is now duplicated for every day the guest stayed.

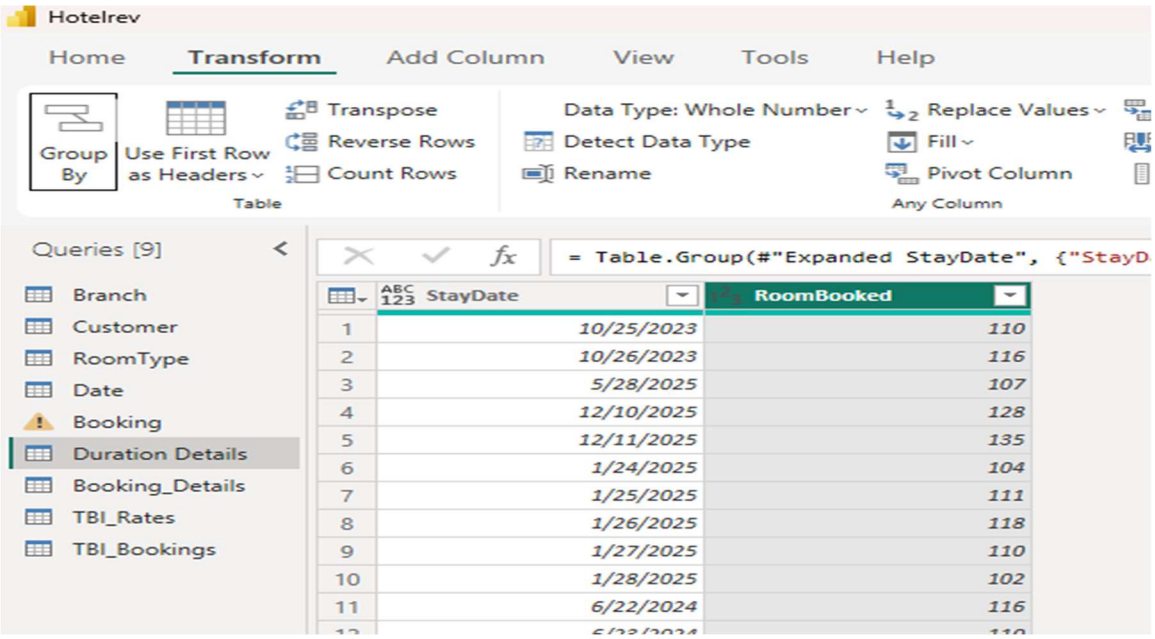
2.1.3 Group By (Occupancy Count):

In the Transform tab, the Group By option was used to group the data by StayDate. A new column named Room Booked was created by counting the number of rows for each date.



2.1.4 Load Data:

The transformed data was loaded into a new worksheet using Home → Close & Load To, and the sheet was renamed as Rooms Booked.



2.2 Creating "Yearly and Monthly Date" (Pivot Table)

This sheet is used for a quick summary view of room bookings aggregated by time.

1. A PivotTable was inserted from the **Rooms Booked** table.
2. Years, Months, and StayDate were added to the **Rows** section.
3. Room Booked was added to the **Values** section with Sum aggregation.
4. The worksheet was renamed **Yearly and Monthly Date**.

Row Labels	Sum of RoomBooked
2023	13887
Qtr3	2930
Qtr4	10957
2024	42412
Qtr1	10465
Qtr2	10735
Qtr3	10708
Qtr4	10504
2025	42152
Qtr1	10035
Qtr2	10720
Qtr3	10619
Qtr4	10778
2026	5753
Qtr1	5753
Grand Total	104204

2.3 Creating "Master Data" Table

1. Create Table: Create a new sheet named "Master Data".
2. Columns & Formulas:

    - Month: Enter Month Starts from 01-Jan-2022 to 31-Dec-2024.
    - Day Count =DAY(EOMONTH(A2,0))
    - Room Available =B2*150
    - Rooms Sold: Take Data from Yearly and Monthly Date table.
    - Occupancy Rate =(D2/C2)*100

MonthStart	Month	DayCount	RoomAvailableCount	RoomSoldCount	Occupancy %
9/1/2023	01-Sep-23	30	9000	2930	32.56%
10/1/2023	01-Oct-23	31	9300	3522	37.87%
11/1/2023	01-Nov-23	30	9000	3759	41.77%
12/1/2023	01-Dec-23	31	9300	3676	39.53%
1/1/2024	01-Jan-24	31	9300	3661	39.37%
2/1/2024	01-Feb-24	29	8700	3397	39.05%
3/1/2024	01-Mar-24	31	9300	3407	36.63%
4/1/2024	01-Apr-24	30	9000	3736	41.51%
5/1/2024	01-May-24	31	9300	3518	37.83%
6/1/2024	01-Jun-24	30	9000	3481	38.68%
7/1/2024	01-Jul-24	31	9300	3741	40.23%
8/1/2024	01-Aug-24	31	9300	3560	38.28%
9/1/2024	01-Sep-24	30	9000	3407	37.86%
10/1/2024	01-Oct-24	31	9300	3536	38.02%
11/1/2024	01-Nov-24	30	9000	3502	38.91%

2.4 Creating Booking_Details Table

1. Create a new Sheet named “Booking_Details”.
2. **Columns and Formulas:**
    - StayDate and Room Booked: Copy from Rooms Booked Table.
    - Month Year =TEXT(A2, "mmm yyyy")
    - Month Start =DATE(YEAR(A2), MONTH(A2), 1)

- Year =YEAR(A2)
- Month =MONTH(A2)
- Season
  =IF(OR(MONTH(A2)=12,MONTH(A2)<=2),"Winter",
  IF(MONTH(A2)<=5,"Spring",
  IF(MONTH(A2)<=8,"Summer","Autumn")))
- Weekday =TEXT(A2,"dddd")
- WeekNum =WEEKNUM(A2,1)
- WeekStart =A2-WEEKDAY(A2)+1
- YearWeek=YEAR(A1+3-WEEKDAY(A1,3)) & "-W" & TEXT(ISOWEEKNUM(A1), "00")

StayDate	RoomBooked	MonthYear	MonthStart	Year	Month	Season	Weekday	WeekNum	WeekStart	YearWeek
10/25/2023	110	10/25/2023	10/1/2023	2023	October	Autumn	Wednesday	43	10/23/2023	2023-W43
10/26/2023	116	10/26/2023	10/1/2023	2023	October	Autumn	Thursday	43	10/23/2023	2023-W43
5/28/2025	107	5/28/2025	5/1/2025	2025	May	Spring	Wednesday	22	5/26/2025	2025-W22
12/10/2025	128	12/10/2025	12/1/2025	2025	December	Winter	Wednesday	50	12/8/2025	2025-W50
12/11/2025	135	12/11/2025	12/1/2025	2025	December	Winter	Thursday	50	12/8/2025	2025-W50
1/24/2025	104	1/24/2025	1/1/2025	2025	January	Winter	Friday	4	1/20/2025	2025-W4
1/25/2025	111	1/25/2025	1/1/2025	2025	January	Winter	Saturday	4	1/20/2025	2025-W4
1/26/2025	118	1/26/2025	1/1/2025	2025	January	Winter	Sunday	5	1/20/2025	2025-W5
1/27/2025	110	1/27/2025	1/1/2025	2025	January	Winter	Monday	5	1/27/2025	2025-W5
1/28/2025	102	1/28/2025	1/1/2025	2025	January	Winter	Tuesday	5	1/27/2025	2025-W5
6/22/2024	116	6/22/2024	6/1/2024	2024	June	Summer	Saturday	25	6/17/2024	2024-W25
6/23/2024	110	6/23/2024	6/1/2024	2024	June	Summer	Sunday	26	6/17/2024	2024-W26
5/28/2024	112	5/28/2024	5/1/2024	2024	May	Spring	Tuesday	22	5/27/2024	2024-W22
5/29/2024	126	5/29/2024	5/1/2024	2024	May	Spring	Wednesday	22	5/27/2024	2024-W22
5/30/2024	124	5/30/2024	5/1/2024	2024	May	Spring	Thursday	22	5/27/2024	2024-W22
5/31/2024	136	5/31/2024	5/1/2024	2024	May	Spring	Friday	22	5/27/2024	2024-W22

2.5 Creating TblRates Table

Create a new sheet named TblRates with a table containing RoomID and Price as columns

RoomTypeID	Price
RT1	2500
RT2	4000
RT3	6000
RT4	9000
RT5	7500
RT6	8000
RT7	1800
RT8	7000
RT9	12000
RT10	5000
RT11	20000

2.6 Creating TblBookings Table

- Create a new sheet.
- Create a Table with Columns BookingID, CustomerID, BranchID, RoomID, CheckInDate, CheckOutDate, NightsStayed, StayDate from Booking_Details Table
- Go to Data →Get Data →Combine Queries → Merge.

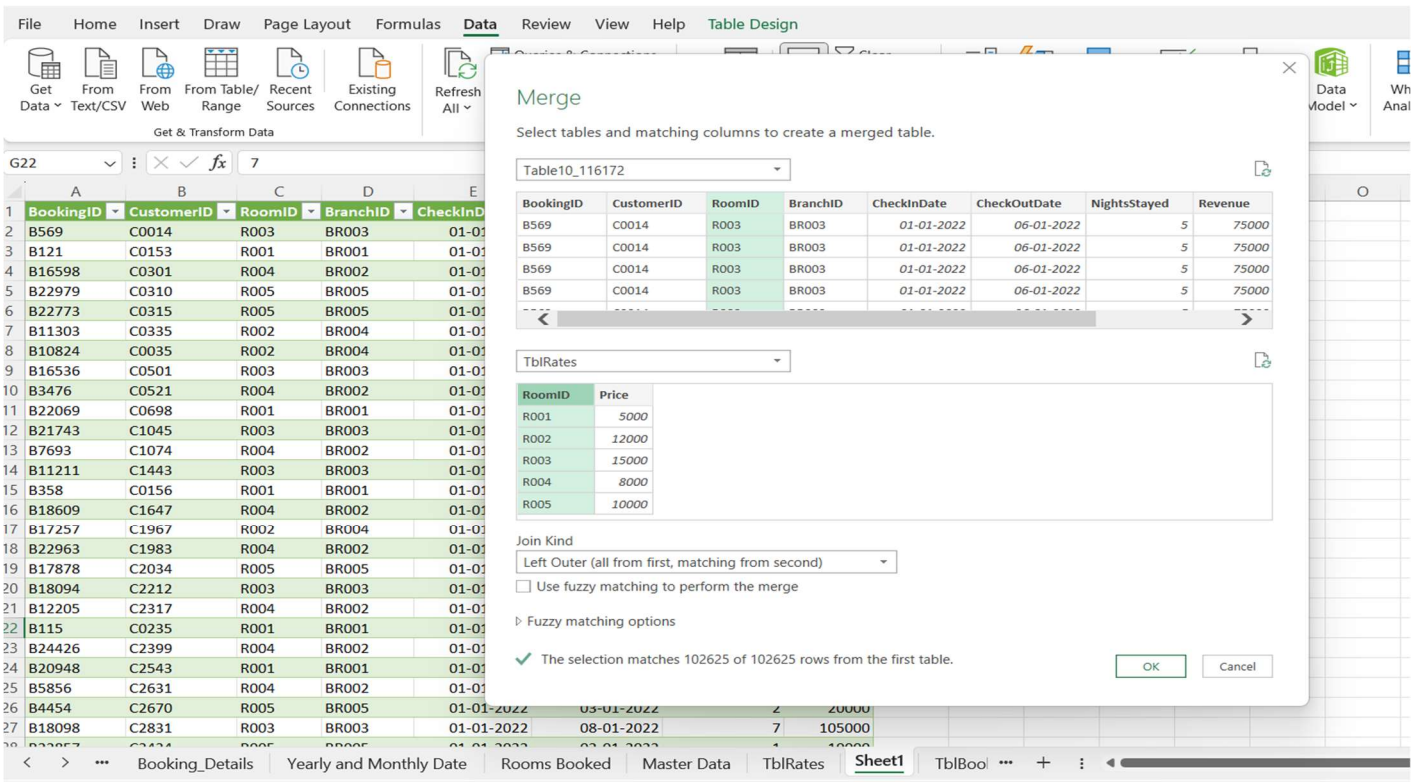
BookingID	DateID	CustomerID	RoomTypeID	BranchID	CheckInDate	CheckOutDate	NightStayed	StayDate	Revenue
BK1000	D001	C2587	RT9	B14	10/25/2023	10/27/2023	2	45224	24000
BK1001	D002	C2700	RT1	B1	5/28/2025	5/29/2025	1	45225	2500
BK1002	D003	C3158	RT10	B1	12/10/2025	12/12/2025	2	45805	10000
BK1004	D005	C1997	RT6	B7	6/22/2024	6/24/2024	2	46001	16000
BK1006	D007	C4894	RT3	B13	1/27/2026	1/28/2026	1	46002	6000
BK1007	D008	C456	RT8	B3	5/4/2024	5/9/2024	5	45681	35000
BK1008	D009	C3708	RT7	B11	8/22/2025	8/23/2025	1	45682	1800
BK1009	D010	C2101	RT10	B12	6/30/2024	7/7/2024	7	45683	35000
BK1011	D012	C3709	RT9	B3	7/25/2025	7/28/2025	3	45684	36000
BK1012	D013	C4958	RT4	B1	9/27/2023	10/1/2023	4	45685	36000
BK1013	D014	C2753	RT3	B10	5/24/2025	5/31/2025	7	45465	42000
BK1014	D015	C2098	RT4	B5	3/26/2024	3/28/2024	2	45466	18000
BK1016	D017	C2938	RT6	B4	3/21/2024	3/25/2024	4	45440	32000
BK1017	D018	C2976	RT11	B8	2/24/2025	3/3/2025	7	45441	140000
BK1019	D020	C4811	RT5	B8	9/2/2024	9/9/2024	7	45442	52500
BK1020	D021	C742	RT5	B6	9/30/2023	10/6/2023	6	45443	45000
BK1022	D023	C3396	RT2	B13	11/5/2025	11/8/2025	3	46049	12000
BK1023	D024	C4047	RT2	B11	12/1/2024	12/8/2024	7	45416	28000
BK1024	D025	C1014	RT3	B5	6/14/2024	6/15/2024	1	45417	6000

2.7 Creating "Tbl_Calculation" (Final KPI Table)

This is the consolidated table used for final calculations using the derived data.

1. **Create Sheet:** Name it **"Tbl_Calculation"**.
2. **Columns & Formulas:**
    - StayDate and Room Booked Count : Copy from Rooms Booked Table.
    - Total Revenue =SUMIFS(TblBookings!J:J, TblBookings!I:I, Tbl_Calculation!A2)
    - Total Rooms = 150
    - Occupancy % =B2/D2*100
    - ADR =C2/B2 (DataType : Currency -> English (India))
    - RevPAR =(F2 * E2)/100
    - Season
      =IF(OR(MONTH(A2)=12,MONTH(A2)<=2),"Winter",
      IF(MONTH(A2)<=5,"Spring",
      IF(MONTH(A2)<=8,"Summer","Autumn")))

- Weekday =TEXT(A2,"dddd")
- Week =WEEKNUM(A2)
- MonthStart =DATE(YEAR(A2), MONTH(A2), 1)
- Month =TEXT(A2,"MMMM")



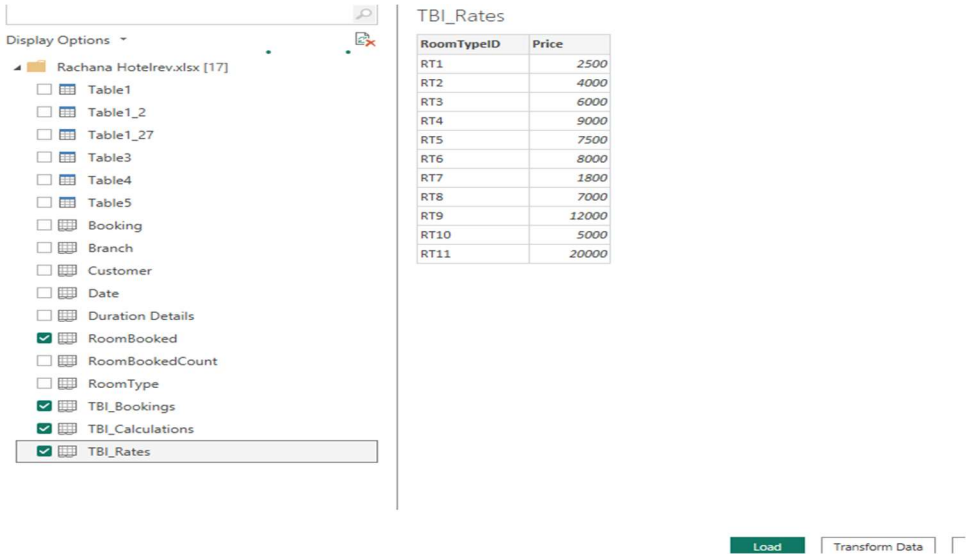
- Quarter ="Q" & ROUNDUP(MONTH(A2)/3, 0)

StayDate	RoomBooked	Total Revenue	Total Rooms	Occupancy	ADR	REVPAR	Season	Weekday	Week	MonthStart	Month	Quarter
10/25/2023	110	24000	300	0.37	₹218.18	₹0.80	Autumn	Wednesday	43	10/1/2023	October	Q4
10/26/2023	116	2500	300	0.39	₹21.55	₹0.08	Autumn	Thursday	43	10/1/2023	October	Q4
5/28/2025	107	10000	300	0.36	₹93.46	₹0.33	Spring	Wednesday	22	5/1/2025	May	Q2
12/10/2025	128	16000	300	0.43	₹125.00	₹0.53	Winter	Wednesday	50	12/1/2025	December	Q4
12/11/2025	135	6000	300	0.45	₹44.44	₹0.20	Winter	Thursday	50	12/1/2025	December	Q4
1/24/2025	104	35000	300	0.35	₹336.54	₹1.17	Winter	Friday	4	1/1/2025	January	Q1
1/25/2025	111	1800	300	0.37	₹16.22	₹0.06	Winter	Saturday	4	1/1/2025	January	Q1
1/26/2025	118	35000	300	0.39	₹296.61	₹1.17	Winter	Sunday	5	1/1/2025	January	Q1
1/27/2025	110	36000	300	0.37	₹327.27	₹1.20	Winter	Monday	5	1/1/2025	January	Q1
1/28/2025	102	36000	300	0.34	₹352.94	₹1.20	Winter	Tuesday	5	1/1/2025	January	Q1
6/22/2024	116	42000	300	0.39	₹362.07	₹1.40	Summer	Saturday	25	6/1/2024	June	Q2
6/23/2024	110	18000	300	0.37	₹163.64	₹0.60	Summer	Sunday	26	6/1/2024	June	Q2
5/28/2024	112	32000	300	0.37	₹285.71	₹1.07	Spring	Tuesday	22	5/1/2024	May	Q2
5/29/2024	126	140000	300	0.42	₹1,111.11	₹4.67	Spring	Wednesday	22	5/1/2024	May	Q2
5/30/2024	124	52500	300	0.41	₹423.39	₹1.75	Spring	Thursday	22	5/1/2024	May	Q2

3. Power BI Data Modelling

3.1 importing Data to PowerBI

Open Power BI Desktop, go to Home → Get Data → Excel Workbook, select the Excel file, choose the required tables (Booking_Details, Rooms Booked, TBI_Rates, TBI_Bookings, TBI_Calculations) in the Navigator, and click Load to import them.



Enhancing Booking_Details (Calculated Columns)

1. Go to the Data View (Table icon on the left).
2. Select the Booking_Details table.
3. Click New Column
4. Formula:

   MonthStart = DATE(YEAR([StayDate]), MONTH([StayDate]), 1)

5. Create Columns in the same process for:

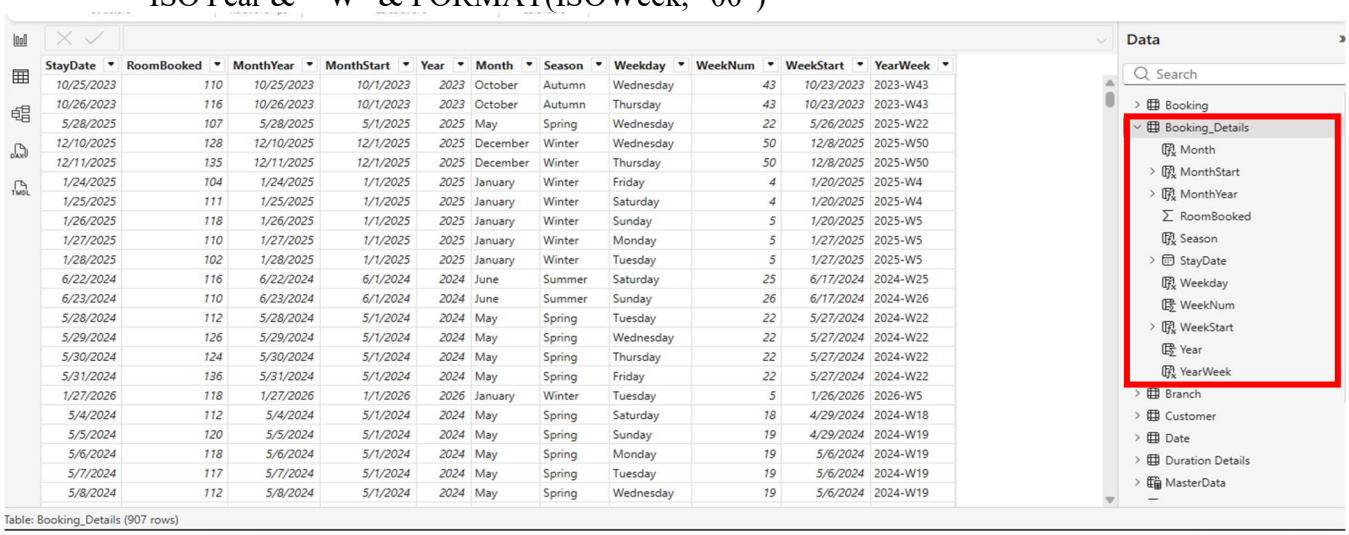
   - MonthYear = FORMAT([StayDate], "MMM YYYY")
   - Year = YEAR([StayDate])
   - Month = FORMAT([StayDate], "MMMM")

   - Season = SWITCH(TRUE(), MONTH([StayDate]) IN {12,1,2}, "Winter",
     MONTH([StayDate]) IN {3,4,5}, "Spring",
     MONTH([StayDate]) IN {6,7,8}, "Summer",
     MONTH([StayDate]) IN {9,10,11}, "Autumn")

   - Weekday = FORMAT([StayDate], "dddd")
   - WeekNum = WEEKNUM([StayDate].[Date])

   - WeekStart = [StayDate] - WEEKDAY([StayDate].[Date]) + 1

   - YearWeek = VAR SundayOfWeek = [StayDate] - WEEKDAY([StayDate], 1) VAR ISOYear = YEAR(SundayOfWeek) VAR ISOWeek = WEEKNUM(SundayOfWeek, 1) RETURN ISOYear & "-W" & FORMAT(ISOWeek, "00")



StayDate	RoomBooked	MonthYear	MonthStart	Year	Month	Season	Weekday	WeekNum	WeekStart	YearWeek
10/25/2023	110	10/25/2023	10/1/2023	2023	October	Autumn	Wednesday	43	10/23/2023	2023-W43
10/26/2023	116	10/26/2023	10/1/2023	2023	October	Autumn	Thursday	43	10/23/2023	2023-W43
5/28/2025	107	5/28/2025	5/1/2025	2025	May	Spring	Wednesday	22	5/26/2025	2025-W22
12/10/2025	128	12/10/2025	12/1/2025	2025	December	Winter	Wednesday	50	12/8/2025	2025-W50
12/11/2025	135	12/11/2025	12/1/2025	2025	December	Winter	Thursday	50	12/8/2025	2025-W50
1/24/2025	104	1/24/2025	1/1/2025	2025	January	Winter	Friday	4	1/20/2025	2025-W4
1/25/2025	111	1/25/2025	1/1/2025	2025	January	Winter	Saturday	4	1/20/2025	2025-W4
1/26/2025	118	1/26/2025	1/1/2025	2025	January	Winter	Sunday	5	1/20/2025	2025-W5
1/27/2025	110	1/27/2025	1/1/2025	2025	January	Winter	Monday	5	1/27/2025	2025-W5
1/28/2025	102	1/28/2025	1/1/2025	2025	January	Winter	Tuesday	5	1/27/2025	2025-W5
6/22/2024	116	6/22/2024	6/1/2024	2024	June	Summer	Saturday	25	6/17/2024	2024-W25
6/23/2024	110	6/23/2024	6/1/2024	2024	June	Summer	Sunday	26	6/17/2024	2024-W26
5/28/2024	112	5/28/2024	5/1/2024	2024	May	Spring	Tuesday	22	5/27/2024	2024-W22
5/29/2024	126	5/29/2024	5/1/2024	2024	May	Spring	Wednesday	22	5/27/2024	2024-W22
5/30/2024	124	5/30/2024	5/1/2024	2024	May	Spring	Thursday	22	5/27/2024	2024-W22
5/31/2024	136	5/31/2024	5/1/2024	2024	May	Spring	Friday	22	5/27/2024	2024-W22
1/27/2026	118	1/27/2026	1/1/2026	2026	January	Winter	Tuesday	5	1/26/2026	2026-W5
5/4/2024	112	5/4/2024	5/1/2024	2024	May	Spring	Saturday	18	4/29/2024	2024-W18
5/5/2024	120	5/5/2024	5/1/2024	2024	May	Spring	Sunday	19	4/29/2024	2024-W19
5/6/2024	118	5/6/2024	5/1/2024	2024	May	Spring	Monday	19	5/6/2024	2024-W19
5/7/2024	117	5/7/2024	5/1/2024	2024	May	Spring	Tuesday	19	5/6/2024	2024-W19
5/8/2024	112	5/8/2024	5/1/2024	2024	May	Spring	Wednesday	19	5/6/2024	2024-W19

3.3 Creating the MasterData Table (DAX Calculated Table)

1. Go to the **Modeling** tab → click **New Table**.
2. **Paste this formula**:

```
MasterData =
VAR CalendarTable =
    ADDCOLUMNS (
        CALENDAR(DATE(2023,9,1), DATE(2026,2,18)),
        "MonthStart", EOMONTH([Date], -1) + 1,
        "Month", FORMAT(EOMONTH([Date], -1) + 1, "dd-mmm-yy"),
        "DayCount", DAY(EOMONTH([Date], 0)),
        "RoomAvailableCount", DAY(EOMONTH([Date], 0)) * 300
    )
RETURN
    ADDCOLUMNS (
        SUMMARIZE (
            CalendarTable,
            [MonthStart], [Month], [DayCount], [RoomAvailableCount]
        ),
        "RoomSoldCount",
            CALCULATE (
                SUM (Booking_Details[RoomBooked]),
                TREATAS ( { [MonthStart] }, Booking_Details[MonthStart] )
            ),
        "Occupancy %",
            DIVIDE (
                CALCULATE (
                    SUM (Booking_Details[RoomBooked]),
                    TREATAS ( { [MonthStart] }, Booking_Details[MonthStart] )
                ),
                [RoomAvailableCount],
                0
            )
    )
```

This will create column MonthStart,Month,Daycount,RoomAvailableCount,Roomsoldcount,Occupancy.

MonthStart	Month	DayCount	RoomAvailableCount	RoomSoldCount	Occupancy %	TotalRevenue	ADR	RevPAR
9/1/2023	01-Sep-23	30	9000	2930	32.56%	₹23,816,900.00	₹8,128.63	₹26.46
10/1/2023	01-Oct-23	31	9300	3522	37.87%	₹26,619,300.00	₹7,558.01	₹28.62
11/1/2023	01-Nov-23	30	9000	3759	41.77%	₹27,795,300.00	₹7,394.33	₹30.88
12/1/2023	01-Dec-23	31	9300	3676	39.53%	₹26,188,000.00	₹7,124.05	₹28.16
1/1/2024	01-Jan-24	31	9300	3661	39.37%	₹27,424,100.00	₹7,490.88	₹29.49
2/1/2024	01-Feb-24	29	8700	3397	39.05%	₹27,503,800.00	₹8,096.50	₹31.61
3/1/2024	01-Mar-24	31	9300	3407	36.63%	₹26,772,700.00	₹7,858.14	₹28.79
4/1/2024	01-Apr-24	30	9000	3736	41.51%	₹28,040,200.00	₹7,505.41	₹31.16
5/1/2024	01-May-24	31	9300	3518	37.83%	₹26,267,000.00	₹7,466.46	₹28.24
6/1/2024	01-Jun-24	30	9000	3481	38.68%	₹25,821,400.00	₹7,417.81	₹28.69
7/1/2024	01-Jul-24	31	9300	3741	40.23%	₹28,578,700.00	₹7,639.32	₹30.73
8/1/2024	01-Aug-24	31	9300	3560	38.28%	₹26,647,000.00	₹7,485.11	₹28.65
9/1/2024	01-Sep-24	30	9000	3407	37.86%	₹24,614,500.00	₹7,224.68	₹27.35
10/1/2024	01-Oct-24	31	9300	3536	38.02%	₹28,074,100.00	₹7,939.51	₹30.19
11/1/2024	01-Nov-24	30	9000	3502	38.91%	₹24,598,600.00	₹7,024.16	₹27.33
12/1/2024	01-Dec-24	31	9300	3466	37.27%	₹27,542,600.00	₹7,946.51	₹29.62
1/1/2025	01-Jan-25	31	9300	3586	38.56%	₹26,574,300.00	₹7,410.57	₹28.57
2/1/2025	01-Feb-25	28	8400	3145	37.44%	₹24,243,000.00	₹7,708.43	₹28.86
3/1/2025	01-Mar-25	31	9300	3304	35.53%	₹25,501,000.00	₹7,718.22	₹27.42
4/1/2025	01-Apr-25	30	9000	3460	38.44%	₹25,489,700.00	₹7,366.97	₹28.32
5/1/2025	01-May-25	31	9300	3674	39.51%	₹28,528,500.00	₹7,764.07	₹30.68

3. Create new columns like TotalRevenue,ADR,REVPAR in master Data Table

Formula:

→ **TotalRevenue** = CALCULATE(
SUM(Bookings[Revenue]),
FILTER(Bookings, Bookings[MonthStart] = MasterData[MonthStart])
)

TotalRevenue:
Calculates the **total revenue generated from bookings** for a specific month by summing the Revenue column based on the matching MonthStart.

→ **ADR** = DIVIDE(

[TotalRevenue],
MasterData[RoomSoldCount]
)

ADR (Average Daily Rate):
Measures the **average revenue earned per sold room**, calculated by dividing TotalRevenue by the number of rooms sold.

→ **RevPAR** = [ADR] * DIVIDE(MasterData[Occupancy %], 100)

RevPAR (Revenue Per Available Room):
Indicates the **revenue generated per available room**, calculated by multiplying ADR with the occupancy rate.

4. Click on New Measure and create the following measures:

DAX MEASURES:

- **MADR** =

```
DIVIDE(
    SUM(Booking[Revenue]),
    SUM(Booking[RoomNightSold]))
```

- **MRevPAR** =

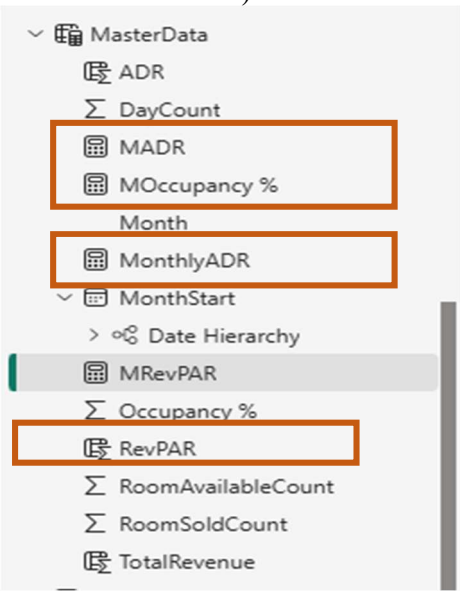
```
DIVIDE(
    SUM(Booking[Revenue]),
    SUM(MasterData[RoomAvailableCount]))
```

- **MOccupancy %** =

```
DIVIDE(
    SUM(Booking[RoomNightSold]),
    SUM(MasterData[RoomAvailableCount]),
    0
) * 100
```

- **MonthlyADR** =

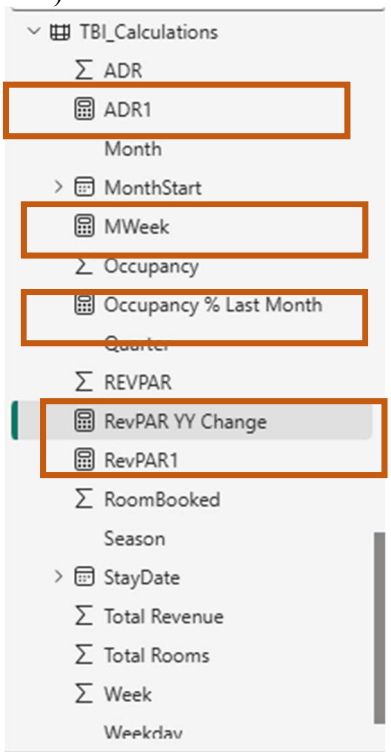
```
MonthlyADR =
CALCULATE(
DIVIDE(SUM(Booking[Revenue]), SUM(MasterData[RoomSoldCount])),
ALLEXCEPT(MasterData, MasterData[MonthStart])
)
```



3.4 Enhancing Tbl_Calculation table

1. Go to the Modeling tab → click New Measure.
2. Create the following measures:

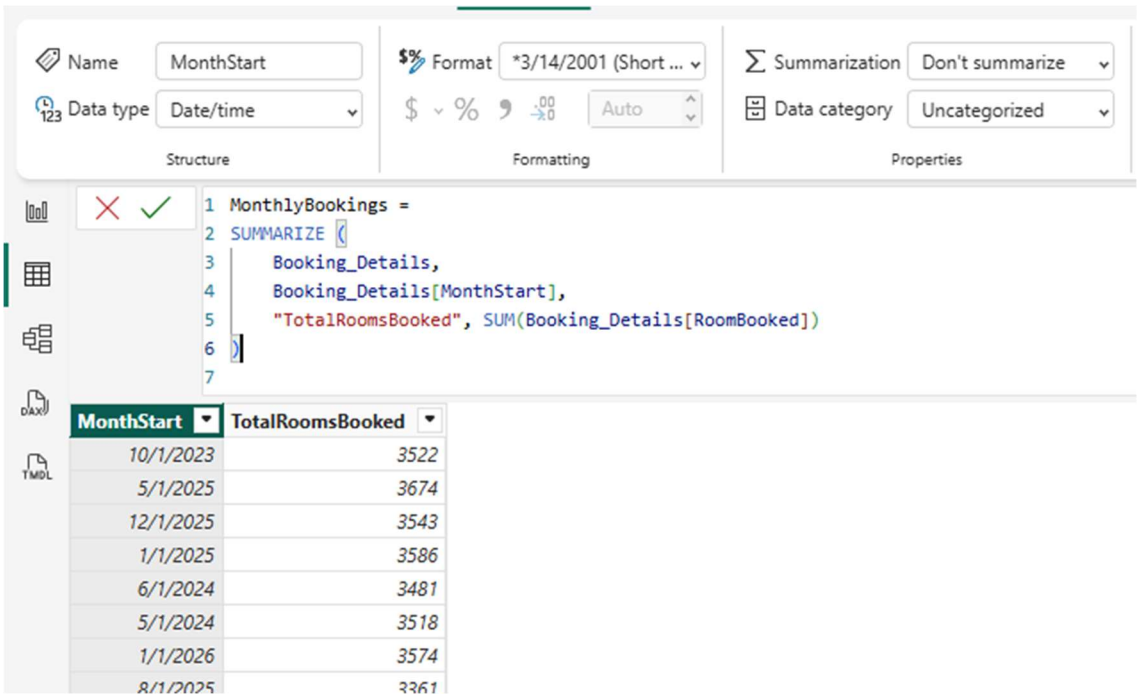
```
ADR YY Change =
    DIVIDE(
        [ADR] - CALCULATE([ADR], SAMEPERIODLASTYEAR('TBI_Calculations'[StayDate])),
        CALCULATE([ADR], SAMEPERIODLASTYEAR('TBI_Calculations'[ [StayDate].[Date]))
    )

MWeek = WEEKNUM(SELECTEDVALUE(TBI_Calculations[StayDate]), 2)


    Occupancy % Last Month = CALCULATE([MOccupancy %],
    PREVIOUSMONTH(TBI_Calculations[ [StayDate]))



    RevPAR YY Change =
    DIVIDE(
        [RevPAR] - CALCULATE([RevPAR], SAMEPERIODLASTYEAR(TBI_Calculations[StayDate])),
        CALCULATE([RevPAR], SAMEPERIODLASTYEAR(TBI_Calculations[StayDate]))
    )
```



3.5 Creation of Monthly Bookings Table

1. Go to the Modeling tab → click New Table.
2. Formula:

```
MonthlyBookings =
SUMMARIZE (
    Booking_Details,
    Booking_Details[MonthStart],
    "TotalRoomsBooked", SUM(Booking_Details[Room Booked])
)
```



4. PowerBI Visuals:

Visual 1: Summary Table

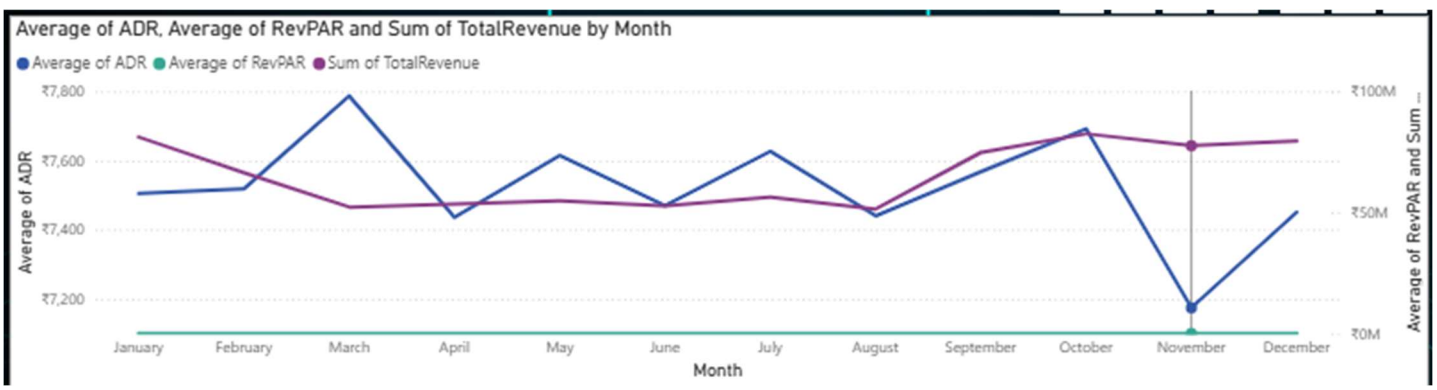
1. Select Visual: Click the Table icon.
2. Add Columns: Drag the following fields in this order:
    - MonthStart
    - RoomAvailableCount (Set to Sum)
    - RoomSoldCount (Set to Sum)
    - TotalRevenue (Set to Sum)
    - ADR (Set to Average)
    - Occupancy % (Set to Average)
    - RevPAR (Set to Average)

Year	Quarter	Month	Day	Sum of RoomAvailableCount	Sum of RoomSoldCount	Sum of TotalRevenue	Average of ADR	Average of RevPAR	Average of Occupancy %
2023	Qtr 3	September	1	9000	2930	₹23,816,900.00	₹8,128.63	₹26.46	32.56%
2023	Qtr 4	October	1	9300	3522	₹26,619,300.00	₹7,558.01	₹28.62	37.87%
2023	Qtr 4	November	1	9000	3759	₹27,795,300.00	₹7,394.33	₹30.88	41.77%
2023	Qtr 4	December	1	9300	3676	₹26,188,000.00	₹7,124.05	₹28.16	39.53%
2024	Qtr 1	January	1	9300	3661	₹27,424,100.00	₹7,490.88	₹29.49	39.37%
2024	Qtr 1	February	1	8700	3397	₹27,503,800.00	₹8,096.50	₹31.61	39.05%
2024	Qtr 1	March	1	9300	3407	₹26,772,700.00	₹7,858.14	₹28.79	36.63%
2024	Qtr 2	April	1	9000	3736	₹28,040,200.00	₹7,505.41	₹31.16	41.51%
2024	Qtr 2	May	1	9300	3518	₹26,267,000.00	₹7,466.46	₹28.24	37.83%
Total				**273600**	**104204**	**₹783,672,300.00**	**₹7,516.19**	**₹28.62**	**38.06%**

Visual 2: Line Chart

1. Select Visual: Click the Line Chart icon.
2. X-Axis: Drag MonthStart from the MasterData table.
3. Y-Axis: Drag the following three fields:
    - ADR (Set to Average)
    - Occupancy % (Set to Average)

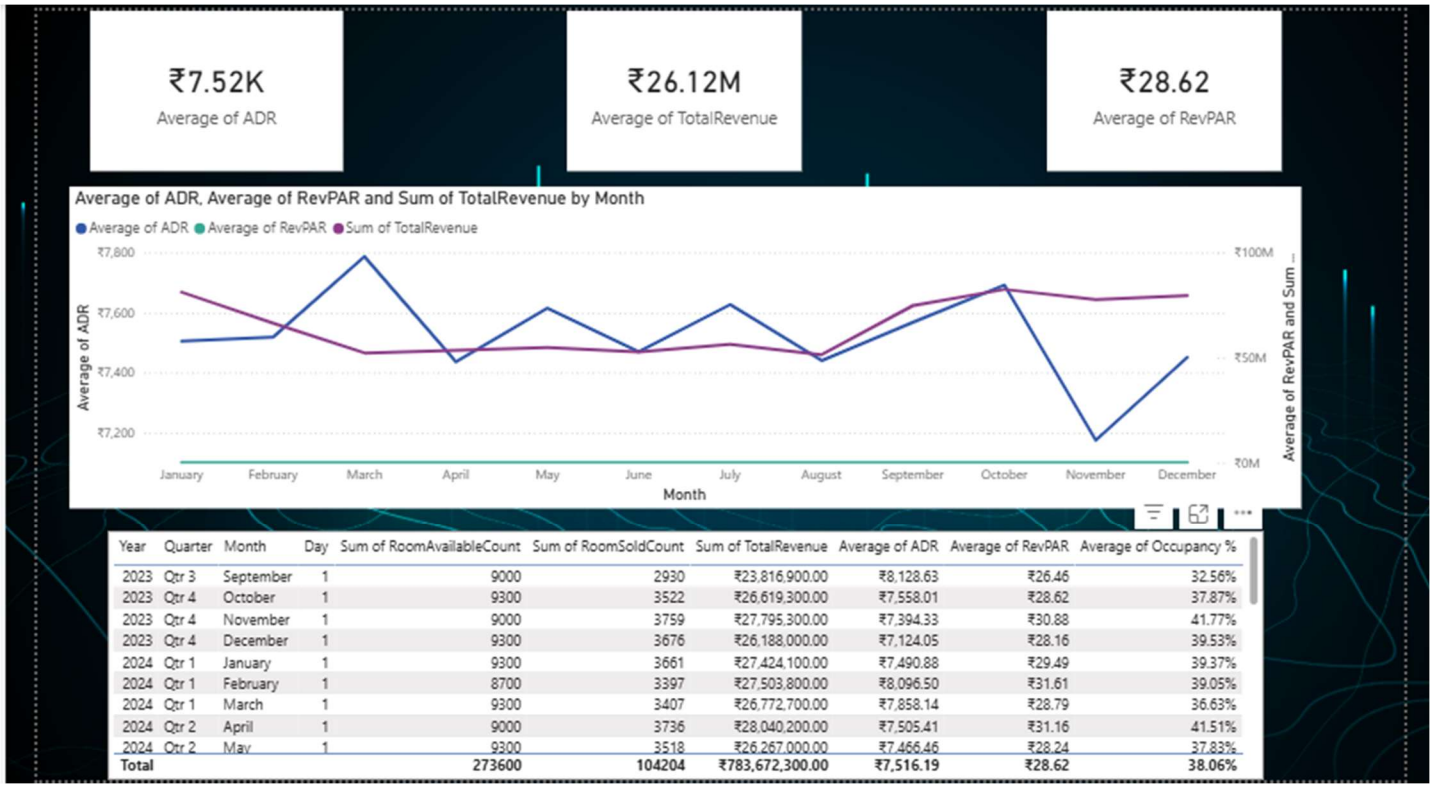
Secondary Y-Axis: RevPAR (Set to Average)



Visual 3: KPI Cards

1. Select Visual: Click the Card visual icon (looks like 123).
2. Card 1 (ADR):
    - Drag ADR from Tbl_Calculation (or your Measure table) to Fields.
    - Ensure aggregation is set to Average (Click dropdown arrow > Average).
    - Format: General > Title > "Average of ADR". Callout Value > Display units > Auto (shows ₹10.00K).
3. Card 2 (RevPAR):
    - Copy/Paste Card 1.
    - Replace field with RevPAR. Set aggregation to Average.

4. Card 3 (Occupancy):
    - Copy/Paste Card 1.
    - Replace field with Occupancy %. Set aggregation to Average.

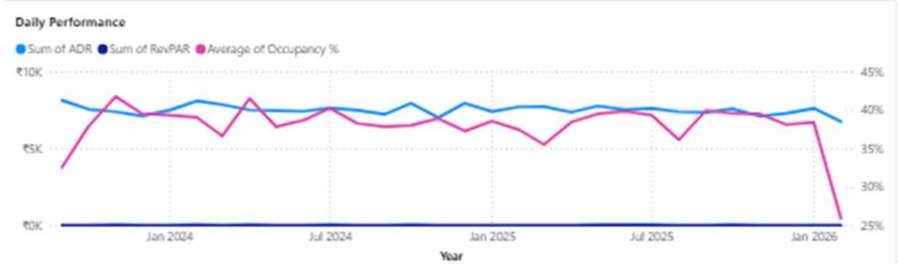


Year	Quarter	Month	Day	Sum of RoomAvailableCount	Sum of RoomSoldCount	Sum of TotalRevenue	Average of ADR	Average of RevPAR	Average of Occupancy %
2023	Qtr 3	September	1	9000	2930	₹23,816,900.00	₹8,128.63	₹26.46	32.56%
2023	Qtr 4	October	1	9300	3522	₹26,619,300.00	₹7,558.01	₹28.62	37.87%
2023	Qtr 4	November	1	9000	3759	₹27,795,300.00	₹7,394.33	₹30.88	41.77%
2023	Qtr 4	December	1	9300	3676	₹26,188,000.00	₹7,124.05	₹28.16	39.53%
2024	Qtr 1	January	1	9300	3661	₹27,424,100.00	₹7,490.88	₹29.49	39.37%
2024	Qtr 1	February	1	8700	3397	₹27,503,800.00	₹8,096.50	₹31.61	39.05%
2024	Qtr 1	March	1	9300	3407	₹26,772,700.00	₹7,858.14	₹28.79	36.63%
2024	Qtr 2	April	1	9000	3736	₹28,040,200.00	₹7,505.41	₹31.16	41.51%
2024	Qtr 2	May	1	9300	3518	₹26,267,000.00	₹7,466.46	₹28.24	37.83%
Total				**273600**	**104204**	**₹783,672,300.00**	**₹7,516.19**	**₹28.62**	**38.06%**

Visual 4: Daily Performance

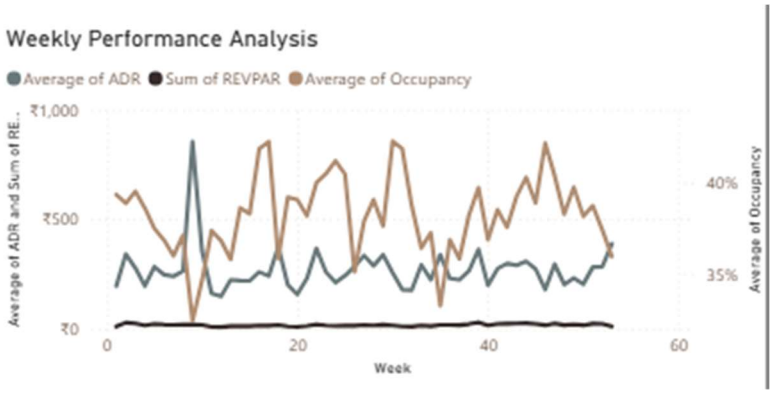
1. Select Visual: Click the Line Chart icon.
2. X-Axis: Drag Year from MasterData.
3. Y-Axis (Left Scale):
    - Drag ADR (Set to Sum)
    - Drag RevPAR (Set to Sum).
4. Secondary Y-Axis (Right Scale):
    - Drag Occupancy % (Set to Average).

5. Go to Format (Paintbrush Icon) -> General -> Title -> Enter “Daily Performance”



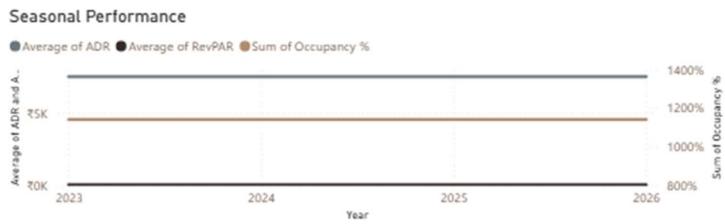
Visual 5: Weekly Performance

1. Select Visual: Click the Line Chart icon.
2. X-Axis: Drag Week from MasterData.
3. Y-Axis: Drag ADR (Average) and RevPAR (Sum).
4. Secondary Y-Axis: Drag Occupancy % (Average).



Visual 6: Seasonal Performance

1. Select Visual: Click the Line Chart icon.
2. X-Axis: Create a hierarchy or drag fields in this order:
    - StayDate (Year, Quarter from Tbl_Calculation)
    - Season (from Tbl_Calculation)
3. Y-Axis: Drag ADR and RevPAR (Averages from MasterData).
4. Secondary Y-Axis: Drag Occupancy % from MasterData (Average).



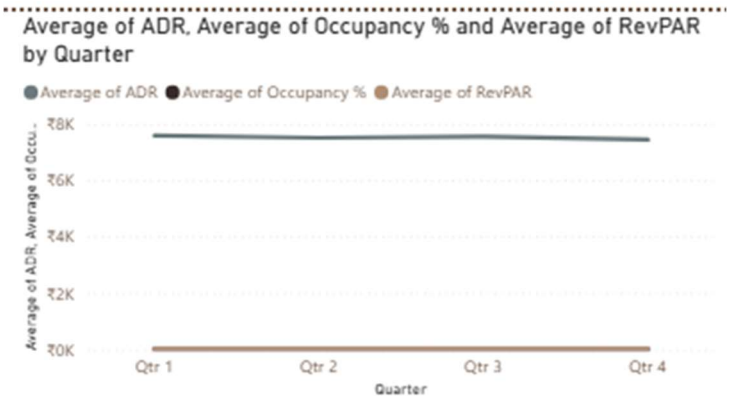
Visual 7: Table

1. Select Visual: Click the Table icon
2. Rows (The Hierarchy): Drag field into the "Rows"
    - MonthStart from MasterData (Year, Quarter, Month, Day)
3. Values: Drag the metrics you want to measure:
    - ADR (Select Average)
    - Occupancy % (Select Average)
    - RevPAR (Select Average)
4. Formatting:
    - Go to Specific Column formatting to apply Currency (₹) to ADR/RevPAR.

Year	Average of ADR	Average of Occupancy %	Average of RevPAR
2024	₹7,591.21	38.64%	₹29.32
2023	₹7,551.26	37.93%	₹28.53
2025	₹7,485.13	38.50%	₹28.81
2026	₹7,182.36	32.19%	₹23.39
Total	**₹7,516.19**	**38.06%**	**₹28.62**

Visual 8: Line Chart

1. **Select Visual:** Line Chart.
2. **X-Axis:** MonthStart from MasterData (with Year, Quarter, Month, Day).
3. **Y-Axis:** ADR (Avg), RevPAR (Avg), Occupancy % (Avg) from MasterData.



5. Insights – Key Performance Indicator (KPI) Analysis

Based on the aggregated data in KPI cards and summary tables:

- **Premium Market Positioning (ADR):**
  The Average Daily Rate (ADR) is about **₹9,996.68 (~₹10,000)**. This indicates the hotel operates in the **premium/luxury segment** with high pricing per room.
- **Operational Efficiency (Occupancy %):**
  The average **occupancy rate is 62%**, meaning **around 38% of rooms remain unsold** on average.
- **Revenue Reality (RevPAR):**
  The **RevPAR is ₹6,232**, which is lower than the ADR. This shows a **gap between potential revenue and actual earnings**, mainly due to vacant rooms.

6. Conclusion

The analysis shows that the hotel chain has **strong pricing power (high ADR)** but **moderate demand (62% occupancy)**. Improving occupancy could help reduce revenue loss and increase overall profitability.